SLIMDIFFUSE: TOWARDS EFFICIENT DIFFUSION-BASED SPEECH ENHANCEMENT USING SLIMMABLE NETWORKS

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ABSTRACT

Diffusion-based models are emerging in the speech enhancement domain and are achieving state-of-the-art performance across various benchmark datasets. A major downside of diffusion models is that data generation requires many evaluations of a typically large neural network, which results in high overall complexity. In this work, we propose a slimmable diffusion model that employs adaptive network widths throughout the data generation process to reduce computational cost. By using a greedy search algorithm to optimize the network width schedule, our method achieves performance comparable to baseline diffusion models with significantly reduced computational complexity. Notably, our approach reduces the computational complexity by up to 87.5% without a significant drop in objective metrics, such as perceptual evaluation of speech quality (PESQ) and SI-SDR.

Index Terms— Diffusion, slimmable neural network

1. INTRODUCTION

Speech enhancement (SE) aims to improve the quality and intelligibility of speech in noisy and reverberant scenarios [1–3]. Significant progress has been achieved in recent years in SE using deep neural networks (DNNs) [4–9]. Most DNN-based SE techniques follow discriminative training paradigms, which perform well in the moderate-to-high signal-to-noise ratio (SNR) regime; however, they exhibit a drop in performance in low SNR scenarios [10]. For such challenging scenarios, generative models gained popularity lately [11–16], as they promise to reconstruct speech content that might be completely masked by noises or distortions.

Recently, diffusion models (DMs) received a lot of attention as a promising approach to generative SE [15, 17–19]. DMs are typically based on a forward noising process, which turns clean speech data into noise by adding Gaussian noise iteratively. This process is reversed in the backward path by denoising the signal step by step at different noise scales. DMs can be formulated in continuous time with stochastic differential equations (SDEs) [20] or in discrete time as denoising diffusion probabilistic models (DDPMs) [21]. Despite their promising performance, DMs are computationally expensive because multiple reverse steps require repeated evaluation of large DNNs, regardless of the underlying SE task complexity, as in SGMSE+ [15]. This makes it challenging to deploy them in real-time applications.

Practical SE applications require DNNs capable of balancing performance and restrictions imposed by hardware and the application, such as memory, latency, etc. Several DNN complexity reduction techniques have been explored for SE: A classical technique is pruning [22], which eliminates redundant parameters, resulting in more lightweight models. DNN architectures can also be adapted during inference by varying their depth, i.e., by using a subset of layers (e.g., early exiting [23]), or by varying their width, i.e., by using a subset of channels (slimmable neural networks (SNNs)) [24]. SNNs use different subsets of their computational graph to avoid redundant computations during inference, which allows for adjusting computational complexity according to the task at hand. Pruning, on the other hand, would have to maintain either multiple networks or would have to rely on a single DNN for all considered tasks, no matter if they require a high or low computational complexity.

In this work, based on SGMSE+, we show that different DNN complexities are required across diffusion steps for SE. Furthermore, we demonstrate that choosing the appropriate DNN complexity for each diffusion step can significantly reduce overall computational cost without degrading signal quality [25]. Based on these findings, we propose SlimDiffuSE, a slimmable diffusion model (SDM) based on SGMSE+ with adaptive network width, which reduces overall computational cost while keeping the same parameter count and achieving comparable SE performance to the original SGMSE+.

*Andreas Brendel has been supported by the Free State of Bavaria by the DSgenAI project.

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2. PROPOSED METHOD

In the SGMSE+ framework [15], SE is formulated as a conditional diffusion process in the complex short-time Fourier transform (STFT) domain. Let $\mathbf{x}_0, \mathbf{y} \in \mathbb{C}^{F\times K}$ denote a clean speech representation and the corresponding noisy observation sampled from the distributions p_{clean} and p_{noisy}, respectively. Here, F and K represent the number of frequency bins and time frames, respectively. The forward diffusion process is defined by the stochastic differential equation (SDE)

$$\mathrm{d}\mathbf{x}_t = \gamma(\mathbf{y} - \mathbf{x}_t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{w}_t, \tag{1}$$

with drift coefficient $\gamma > 0$, time-dependent diffusion coefficient $g(t) > 0$ and standard complex-valued Wiener process $\mathbf{w}_t$. This process gradually perturbs $\mathbf{x}_0$ toward $\mathbf{y}$ by adding (Gaussian) diffusion noise scaled by $g(t)$, where $\mathbf{x}_t \in \mathbb{C}^{F\times K}$ is the noisy signal at diffusion time t. The corresponding reverse-time SDE used for inference is given by

$$\mathrm{d}\mathbf{x}_t = -\gamma(\mathbf{y}-\mathbf{x}_t)\,\mathrm{d}t + g^2(t)\,\mathbf{S}_\theta(\mathbf{x}_t, \mathbf{y}, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\bar{\mathbf{w}}_t, \tag{2}$$

where $\bar{\mathbf{w}}_t$ denotes a reverse-time Wiener process and $\mathbf{S}_\theta(\mathbf{x}_t, \mathbf{y}, t)$ is a DNN parameterized by θ that estimates the conditional score $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathbf{y})$. In practice, inference is performed using a Predictor–Corrector (PC) sampler with N discretization steps [20], resulting in the following computational cost

$$\mathcal{C}_{\text{total}} = 2N \cdot \text{FLOPs}(\mathbf{S}_\theta), \tag{3}$$

where the factor 2 accounts for the predictor and corrector updates at each diffusion step, and FLOPs($\mathbf{S}_\theta$) quantifies the computational complexity of the score model in terms of floating point operations (FLOPs). While effective, this formulation allocates a fixed network capacity at every diffusion step, regardless of the actual difficulty of the denoising task, which results in inefficient computational allocation across all reverse-diffusion steps. We hypothesize that during the reverse diffusion process, early time steps (t close to 1) require score models with high model capacity, as signal structure has to be built up from noise, whereas later time steps (t close to 0) primarily refine fine-grained spectral details and are comparatively less demanding (see Sec. 3.2). Hence, assigning lower model complexities to later diffusion time steps should be sufficient for high-quality data generation.

To address this limitation, we propose SlimDiffuSE, a slimmable score model $\mathbf{F}^u_\phi(\mathbf{x}_t, \mathbf{y}, t)$, where $u \in (0, 1]$ is a continuous utilization factor controlling the active network width. For a given u, the active sub-network is obtained by applying a structured channel-wise mask that retains the first $\lceil u\,C\rceil$ channels in each slimmable layer, where C denotes the full number of channels (see Fig. 1) and $\lceil\cdot\rceil$ is the ceiling operator. The model is trained using a multi-width optimization strategy [24] over a set of utilization factors $\mathcal{U} = \{0.25, 0.5, 1.0\}$

$$\mathcal{L}(\phi) = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{y},\mathbf{z}}\left[\sum_{u\in\mathcal{U}} u \left\| \mathbf{F}^u_\phi(\mathbf{x}_t, \mathbf{y}, t) + \frac{\mathbf{z}}{\sigma(t)} \right\|_2^2\right], \tag{4}$$

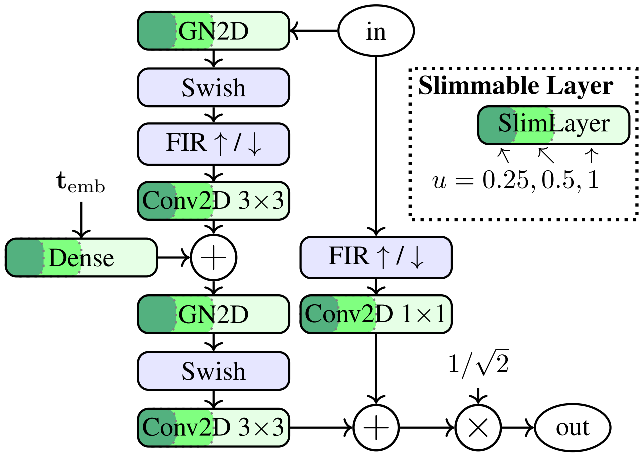

Fig. 1. Residual block with slimmable channels.

where $\mathbf{z} \sim \mathcal{N}_\mathbb{C}(\mathbf{0}, \mathbf{I})$ and $\sigma(t)$ denotes the variance of the diffusion noise schedule. During inference, we perform N steps in the reverse process and apply a step-dependent utilization schedule $u_n \in \mathcal{U}$, $n = 1, \dots, N$. The update rule for the reverse diffusion process is obtained as

$$\begin{aligned} \mathbf{x}_{n-1} &= \mathbf{x}_n - \gamma(\mathbf{y} - \mathbf{x}_n)\Delta t \\ &\quad + g^2(t_n)\,\mathbf{F}^{u_n}_\phi(\mathbf{x}_n, \mathbf{y}, t_n)\Delta t + g(t_n)\sqrt{\Delta t}\,\mathbf{z}_n, \end{aligned} \tag{5}$$

where $\mathbf{z}_n \sim \mathcal{N}_\mathbb{C}(\mathbf{0}, \mathbf{I})$. This adaptive formulation enables dynamic allocation of model capacity across inference steps. Assuming the computational complexity scales linearly with the utilization factor u, the resulting computational cost is approximated as:

$$\mathcal{C}_{\text{slim}} = 2\sum_{n=1}^{N} \text{FLOPs}(\mathbf{F}^{u_n}_\phi) \approx 2\,\text{FLOPs}(\mathbf{S}_\theta) \cdot \sum_{n=1}^{N} u_n \tag{6}$$

which can be optimized by parameterizing the slimmable score model $\mathbf{F}^u_\phi$, $u < 1$, with lower complexity to later diffusion steps (n higher), aiming at fine structure, while assigning higher model capacity for the early stages of the reverse diffusion process (n small).

SlimDiffuSE architecture: In this work, we adapt the NCSN++ backbone [26], a multi-resolution U-Net with skip connections and a parallel-growing U-Net path, which is used in SGMSE+ [15]. Each upsampling, downsampling, and bottleneck module is realized by residual blocks [27]. We use three residual blocks per upsampling module and two per downsampling module, where the last residual block handles up-/down-sampling. Each residual block consists of convolution layers, group normalization (GN), finite impulse response (FIR) filters for up/down sampling, and Swish activation, as shown in Fig. 1. A global attention module is applied to the bottleneck at a resolution of 16×16. Fourier time embeddings $\mathbf{t}_{\text{emb}} \in \mathbb{R}^{128}$ are fed to each residual block. The number of channels of the network layers is computed by multiplying the base channel number $C_{\text{base}} = 128$ with channel multipliers $(1, 1, 2, 2, 2, 2, 2)$. Different model complexities can be obtained by varying C_{base}. For a slimmable

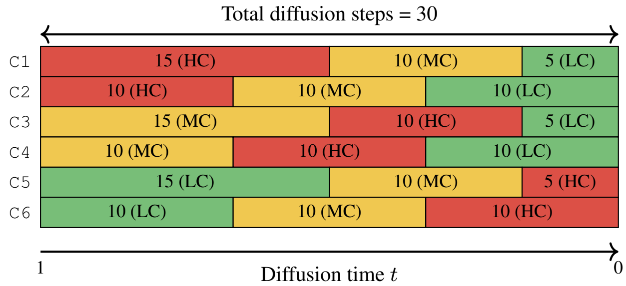

Fig. 2. Network configurations (C1–C6) with three model complexities assigned to different reverse diffusion steps.

Table 1. Objective evaluation metrics for the three baseline complexities (HC, MC, LC) and the six predefined mixed complexity configurations C1-C6.

Model	PESQ ↑	SI-SDR (dB) ↑	ESTOI ↑	SI-SIR (dB) ↑
Noisy	1.58 ± 0.46	9.1 ± 5.5	0.81 ± 0.12	9.1 ± 5.5
HC	**2.85 ± 0.72**	**16.9 ± 6.2**	**0.93 ± 0.07**	**27.4 ± 9.3**
MC	2.72 ± 0.73	15.8 ± 6.2	0.92 ± 0.07	25.9 ± 9.7
LC	2.48 ± 0.72	14.4 ± 6.1	0.90 ± 0.08	23.1 ± 9.4
C1	2.82 ± 0.70	**16.9 ± 6.3**	**0.93 ± 0.07**	**27.6 ± 9.5**
C2	**2.83 ± 0.71**	16.8 ± 6.3	**0.93 ± 0.07**	27.2 ± 9.4
C3	2.72 ± 0.73	15.8 ± 6.3	0.92 ± 0.07	26.2 ± 10.0
C4	2.72 ± 0.74	15.9 ± 6.3	0.92 ± 0.07	26.3 ± 9.7
C5	2.54 ± 0.72	14.8 ± 6.1	0.90 ± 0.08	24.4 ± 9.3
C6	2.56 ± 0.71	14.8 ± 6.0	0.90 ± 0.08	24.1 ± 9.1

architecture, each residual block is converted into a slimmable residual block, where convolutions, GNs, and dense layers adapt their input/output dimensions according to a utilization factor $u \in \mathcal{U} = \{0.25, 0.5, 1\}$ following [24], as shown in Fig. 1. The bottleneck attention module is likewise transformed into a slimmable attention layer controlled by u. This construction results in three effective model widths with $C_{\text{base}} \in \{32, 64, 128\}$.

3. EXPERIMENTS AND RESULTS

3.1. Experimental Details

Datasets: Training data was generated using the Interspeech 2020 DNS Challenge dataset [28] by mixing clean speech with noise recordings at randomly sampled SNRs from $[-10, 30]$ dB following a setup similar to [5, 29]. To simulate reverberation, 50% of the clean utterances were convolved with randomly selected room impulse responses (RIRs) from the DNS RIR dataset [28], which contains approximately 100k measured and synthetic RIRs with reverberation times ranging from 0.3 to 1.5 s. For evaluation, the DNS Challenge non-reverberant test set [28] was used, which includes noises from 12 VoIP-relevant categories, with SNRs in the range $[0, 25]$ dB, and consists of 150 test samples of 10 s each.

Training and evaluation: The baseline SGMSE+ [15] and the proposed models were trained with the Adam optimizer at a learning rate of 10^{-4} using an STFT window size of 510, a hop length of 128, an FFT length of 510, and with a sampling rate of 16kHz. An exponential moving average (EMA) of the DNN weights with a decay factor of 0.999 [30] was calculated, and the best-performing model checkpoints were selected based on the highest validation PESQ scores [31]. We evaluated all methods using the following objective metrics: PESQ [31], extended short-time objective intelligibility (ESTOI) [32], scale-invariant signal-to-distortion ratio (SI-SDR) [33], and scale-invariant signal-to-interference ratio (SI-SIR) [33]. We quantified computational complexity by giga-multiply-accumulate operations per second (GMACs) and the total number of model parameters.

3.2. Experimental Results

Models at different complexity: In this experiment, we show that the difficulty of the iterative denoising task in DMs for SE varies across diffusion time steps. Exploiting this, we reduce model complexity at selected steps while maintaining performance, substantially lowering computational cost. To this end, we trained three score models based on the NCSN++ architecture: high-complexity (HC), medium-complexity (MC), and low-complexity (LC), with base channel dimensions $C_{\text{base}} \in \{128, 64, 32\}$ corresponding to utilization factors $u \in \{1.0, 0.5, 0.25\}$, respectively. We evaluated the assignment of these models to diffusion time steps in six predefined configurations (see Fig. 2). The results in Tab. 1 demonstrate that applying a low-complex model (LC, MC) in all reverse diffusion steps leads to performance degradation: PESQ scores dropped by at least 0.13 and 0.37 for the MC and LC variant, respectively, compared to the HC model. However, among the mixed-complexity configurations, variants C1 and C2 achieved results almost matching the performance of the full HC evaluation. Specifically, C1 and C2 yielded PESQ scores of 2.82 and 2.83, approaching the 2.85 achieved by the HC model, while reducing the overall computational load. These results also indicated that employing more model complexity at the beginning of the reverse diffusion process ($t \to 1$) is important. We observed a notable performance drop for the C3 through C6 configurations, where the HC model is instead employed in the middle or final segments ($t \to 0$). This behavior is intuitive, as at early reverse-time steps ($t \to 1$), the model must build up signal structures from pure noise, requiring higher network capacity. Conversely, as the sample approached the clean speech distribution ($t \to 0$), the denoising task becomes less demanding, and a low-complexity model is sufficient for generation.

Slimmable diffusion model: While our initial experiments validated that predefined combinations of models of varying width could match the performance of the full HC model with significantly reduced computational complexity, this approach increased the overall memory footprint. For example, the C2 configuration (see Tab. 1) reduced complexity by over 56% (from 125 to 54.8 GMACs/Frame), but increased the to-

Table 2. Objective evaluation metrics for slimmable models and their mixed-complexity configurations.

Model	PESQ ↑	SI-SDR (dB) ↑	ESTOI ↑	SI-SIR (dB) ↑
$u = 1.0$	**2.78 ± 0.68**	**16.8 ± 5.7**	**0.92 ± 0.07**	**28.9 ± 9.2**
$u = 0.5$	2.57 ± 0.75	15.6 ± 6.0	0.90 ± 0.08	26.8 ± 9.3
$u = 0.25$	2.18 ± 0.70	13.3 ± 5.8	0.88 ± 0.10	23.1 ± 10.0
`C1`	2.76 ± 0.67	16.7 ± 5.6	**0.92 ± 0.07**	29.2 ± 8.8
`C2`	**2.77 ± 0.67**	**16.8 ± 5.6**	**0.92 ± 0.07**	**29.5 ± 9.1**
`C3`	2.56 ± 0.74	15.6 ± 5.9	0.90 ± 0.08	26.8 ± 8.9
`C4`	2.60 ± 0.74	15.8 ± 6.0	0.91 ± 0.08	27.1 ± 9.0
`C5`	2.28 ± 0.71	14.0 ± 5.7	0.88 ± 0.10	25.1 ± 9.1
`C6`	2.28 ± 0.70	14.0 ± 5.7	0.88 ± 0.09	25.3 ± 9.7

Table 3. Objective performance across search-optimized schedules. Intervals $[s_{\text{start}}, s_{\text{end}}]$ denote reverse sampling steps where specific utilization factors u are active.

Complexity Transitions			Avg. u	GMACs /Frame	PESQ ↑	SI-SDR (dB) ↑
$u = 1.0$	$u = 0.5$	$u = 0.25$				
$[1, 1]$	$[2, 6]$	$[7, 30]$	**0.317**	**15.62**	2.77	16.7
$[1, 2]$	$[3, 7]$	$[8, 30]$	0.342	19.54	2.76	16.6
$[1, 3]$	$[4, 8]$	$[9, 30]$	0.367	23.46	**2.78**	16.7
$[1, 5]$	$[6, 10]$	$[11, 30]$	0.417	31.30	2.76	16.6
$[1, 7]$	$[8, 12]$	$[13, 30]$	0.467	39.14	2.76	16.6
$[1, 10]$	$[11, 20]$	$[21, 30]$	0.542	50.90	2.77	**16.8**
$[1, 15]$	$[16, 25]$	$[26, 30]$	0.667	70.50	2.76	16.7
HC	—	—	1.000	125.40	2.85	16.9
$u = 1.0$	—	—	1.000	125.40	2.78	16.8

tal number of parameters from 65 M to 89 M as three separate score models with 65, 18.1, and 5.5M parameters have to be maintained. In contrast, as outlined in Sec. 2, a slimmable model with variable width adapts its complexity across diffusion steps without incurring the parameter overhead of multiple models. Tab. 2 presents the results for SlimDiffuSE across different utilization factors u for 30 reverse-diffusion steps. The full-width configuration ($u = 1.0$) achieved performance comparable to the HC baseline. However, reduced-width configurations ($u = 0.5, 0.25$) show some degradation compared to the standalone MC and LC models (see Tab. 1), since parameter sharing across widths imposes a joint optimization objective on the shared weights. In particular, PESQ drops to 2.57 and 2.18, compared to 2.72 and 2.48 for MC and LC, respectively. Despite this, when using predefined mixed-complexity configurations (replacing HC, MC, and LC with corresponding SlimDiffuSE configurations), configurations `C1` and `C2` achieved performance comparable to the original mixed-complexity results reported in Tab. 1 for these configurations.

Greedy search: We further aim to determine an optimal allocation of model widths along the reverse-diffusion steps for SlimDiffuSE. To this end, we adopted an evolutionary-inspired greedy search algorithm that operated over the set of utilization factors $\mathcal{U} = \{1.0, 0.5, 0.25\}$ as shown in Alg. 1. Starting from a fully utilized model ($u = 1.0$), the method progressively replaced later diffusion steps with lower utilization factors, while maintaining performance within a tolerance ϵ relative to the baseline ($u = 1.0$). In this work, the tolerance ϵ is based on the objective metric PESQ, calculated on a validation dataset $\mathcal{D}_{\text{val}}$ (a smaller disjoint subset of the training dataset) for a given reverse diffusion schedule S. We enforced a monotonic, non-increasing utilization constraint for decreasing t, which significantly reduced the search space. Thus, instead of evaluating all L^T configurations ($L = |\mathcal{U}|$), the problem reduced to identifying the transition boundaries, resulting in $\mathcal{O}(LT)$ complexity, instead of $\mathcal{O}(L^T)$. For $L = 3$ and $T = 30$, this reduced evaluations from $3^{30} \approx 2.05 \times 10^{14}$ to at most 60. As summarized in Tab. 3, the proposed greedy search strategy identified highly efficient configurations that maintain competitive performance while drastically reducing computational overhead. In particular, an optimized configuration with average utilization of 0.317 and 15.62 GMACs/Frame achieved a PESQ of 2.77, comparable to the HC baseline (2.85) and SlimDiffuSE full-width variant (2.78 for $u = 1.0$). Overall, this corresponded to an 87.5% reduction in computational cost relative to the HC model, demonstrating that competitive performance can be achieved even when most of the denoising trajectory is offloaded to lower-complex sub-networks.

Algorithm 1 Greedy search over utilization factors.

Input: Number of diffusion steps T, utilization factors $\mathcal{U} = \{1.0, 0.5, 0.25\}$, tolerance $\epsilon = 0.1$, validation set $\mathcal{D}_{\text{val}}$
Initialize: schedule $S = [1.0, \dots, 1.0]$
Initialize: set of valid configurations $\mathcal{S}_{\text{all}} = \{S\}$
$P_{\text{base}} \leftarrow \text{PESQ}(S, \mathcal{D}_{\text{val}})$
for $u = 0.5, 0.25$ **do**
  $i \leftarrow 1$
  while $i \leq T$ **do**
    $S' \leftarrow S$
    $S'_{T-i+1:T} = u$
    if $\text{PESQ}(S', \mathcal{D}_{\text{val}}) < P_{\text{base}} - \epsilon$ **then break**
    $S \leftarrow S', i \leftarrow i + 1$
    $\mathcal{S}_{\text{all}} \leftarrow \{S\} \cup \mathcal{S}_{\text{all}}$
  end while
end for
return $\mathcal{S}_{\text{all}}$

4. CONCLUSIONS

In this work, we showed that the difficulty of the per-step denoising task in diffusion-based SE varies along the diffusion trajectory. Our proposed SlimDiffuSE uses a greedy search algorithm for optimizing the slimming schedule and achieves performance comparable to a full high-complexity model while significantly reducing computational cost. We believe our work establishes a flexible trade-off between SE quality and efficiency, paving the way for real-time diffusion-based enhancement on resource-constrained hardware.

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